Stable and Safe Neural ODEs

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Introductions

- Our work explores Neural Ordinary Differential Equations (NODE) to learn generalized Dynamical System (DS) from demonstrated trajectories.
- We learn the NODE using a MLP and a corresponding Lyapunov Function to guarantee stability during training and rollout.
- We evaluate our NODE on a collision avoidance task by applying CBF and CLF constraints to **ensure both stability and safety**.

Framework

1. Neural ODEs

Goal. Learn smooth, time-continuous policies

Model.

$$\dot{x} = f(x), f: \mathbb{R}^2 \to \mathbb{R}^2 (MLP)$$

2. Stability: ICNN Lyapunov + CLF Projection[2]

Goal. Global asymptotic convergence to x^*.

Warped ICNN Lyapunov function:

$$\boldsymbol{V(x) = \sigma_{k+1}\left(g(F(x)) - g(F(0))\right) + \epsilon \parallel x \parallel_2^2}$$

where

- F is a residual warp network
- g is an ICNN acting on the warped coordinates
- σ is the outer smoothed ReLU activation.

CLF condition:

$$\dot{V}(x) + \alpha V(x) \leq 0,\ \alpha > 0$$

Analytic CLF projection:

$$\pi_{clf}(\hat{f}) = \hat{f} - \frac{\max\left(0, \nabla V \cdot \hat{f} + \alpha V\right)}{\parallel \nabla V \parallel^2} \nabla V$$

3. Safety

a. QP-based safety[1]

Idea. At training or inference time, solve a QP:

$$f^*(x) = \arg\min_u \frac{1}{2} \parallel u - f(x) \parallel^2 + p\delta^2$$
$$\text{s.t.} \quad \nabla h_j(x) \cdot u + \alpha_h h_j(x) \geq 0, \forall j \quad (CBF)$$
$$\nabla V(x) \cdot u + \alpha V(x) \leq \delta \quad (CLF)$$
$$\delta \geq 0$$

Neural ODE formulation:

$$\dot{x} = \Sigma_{i=1}^{n_2} \theta_{K-1,K}^i z_{K-1}^i = \theta_{K-1,K}^P z_{K-1}^P + \theta_{K-1,K}^N z_{K-1}^N$$

CBF Constraint in parameter space:

$$\frac{dh}{dx}\theta_{K-1,K}^P z_{K-1}^P + \frac{dh}{dx}\theta_{K-1,K}^N z_{K-1}^N + \alpha h(x) \geq 0$$

CLF Constraint in parameter space:

$$\frac{dV}{dx}\theta_{K-1,K}^P z_{K-1}^P + \frac{dV}{dx}\theta_{K-1,K}^N z_{K-1}^N + \alpha V(x) \geq 0$$

Treat $\theta_{K-1,k}^P$ as virtual control variables

Framework

b. Analytic projection

Idea. Instead of solving QP for safety and stability, enforce feasibility using projection

Bounded elliptic obstacle:

$$h_j(x) = \sum_{n_{obs}} \left(\frac{x_i - c_{j,i}}{r_{j,i}}\right)^2 - 1$$

where $c_{j,i}, r_{j,i}$ are the center and radius of obstacle j.

Sequential analytic projection (POCS):

1. CLF projection: Project onto $\dot{V} + \alpha V \leq 0$ using π_{clf}
2. CBF projection: Project onto $\dot{h} + \beta h \leq 0$ using

$$\pi_{cbf}(\hat{f}) = \hat{f} - \frac{\max\left(0, -(\nabla h \cdot \hat{f} + \alpha h\right)}{\parallel \nabla h \parallel^2} \nabla V$$

c. HJB Reachability Condition[3]

Replace CBF with HJB condition:

$$V(x) = \min_{\tau \in [0, \infty)} \max_u h(x(\tau))$$

Training

Joint training of f_θ and V_ϕ on demonstrations:

$$\mathcal{L} = \mathcal{L}_{task} + \lambda_v \mathcal{L}_{proj}^V + \lambda_H \mathcal{L}_{proj}^h + \lambda_{vc} \mathcal{L}_{vc} + \lambda_{sm} \mathcal{L}_{sm}$$

- $\mathcal{L}_{task}$: $\left\| f_{proj}(x_i) - \dot{x}_i \right\|^2$- imitation through the projection
- $\mathcal{L}_{proj}^V$: $\|\Delta f_{CLF}(\hat{f})\|^2$- shape raw $\hat{f}$ to be CLF-feasible
- $\mathcal{L}_{proj}^h$: $\left\|\Delta f_{CBF}(\hat{f})\right\|^2$- shape raw $\hat{f}$ to be CBF-feasible
- $\mathcal{L}_{vc}$: kNN tangent-velocity copy at off-demo samples
- $\mathcal{L}_{sm}$: $ReLU(V(x_i) + \epsilon \dot{x}_i) - V(x_i))$

Demonstrations: $\{(x_i, \dot{x}_i)\}_{i=1}^N$ with attractor x^*, obstacles $\mathcal{O} = \{(c_j, r_j)\}_{j=1}^M$

Architecture:

- f_θ - NODE: MLP (dim=128) with tanh activation function
- V_{Lyap}- Warped-ICNN (warp_dim=64, ICNN hidden=[64,64],outer ReLU.)

Training & Inference:

To learn QP-based safety, we apply 2-stage training:

1. Train (f, V) jointly without obstacles
2. Fix (f, V) , and train $class\ K$ function for CBF using differentiable QP.

During inference, run the CBF+CLF QP at each step to filter the nominal field.

To learn projection-based safety, we integrate the analytic CLF + CBF projection layer directly into the forward pass and train end-to-end in a single stage.

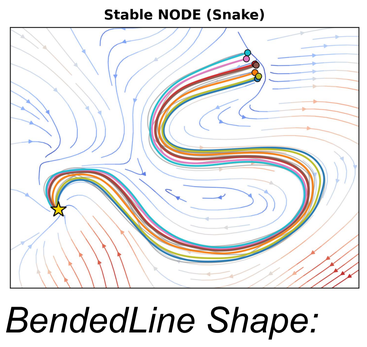
To learn HJB value function, we apply MSE loss.

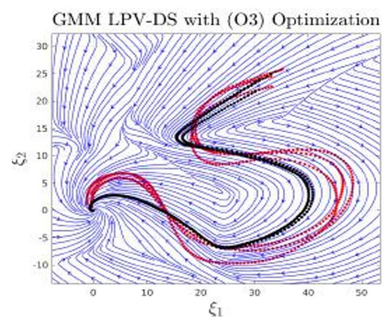
Evaluation

We test the method on LASA handwriting dataset

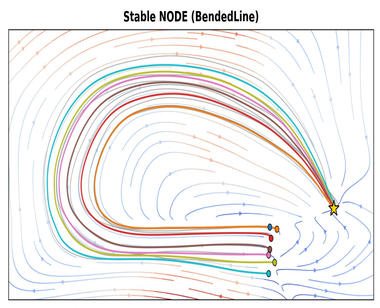
1. NODE performance compared to LPVDS[4]:

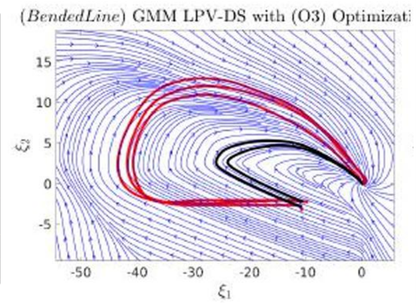
Snake Shape:



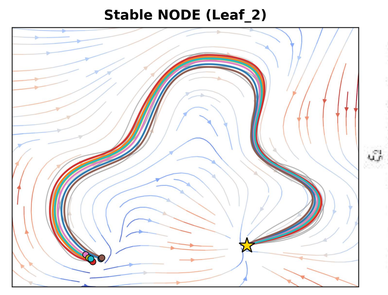


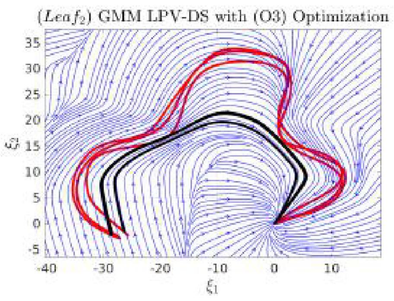
BendedLine Shape:





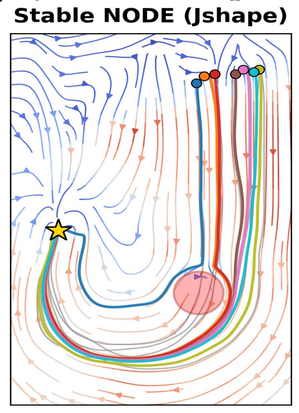
Leaf_2 Shape:

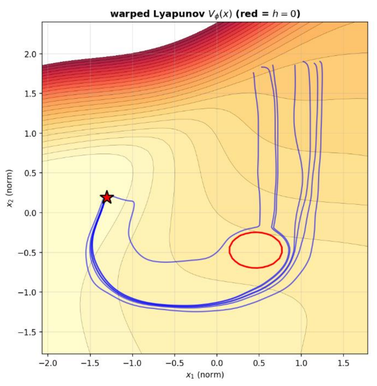


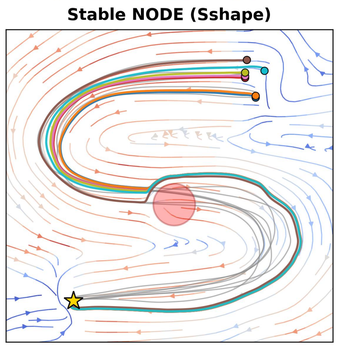


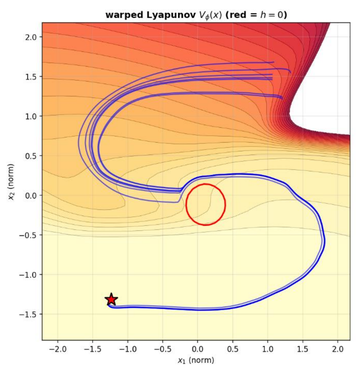
2. Obstacle Avoidance:

Safety Behavior using CBF

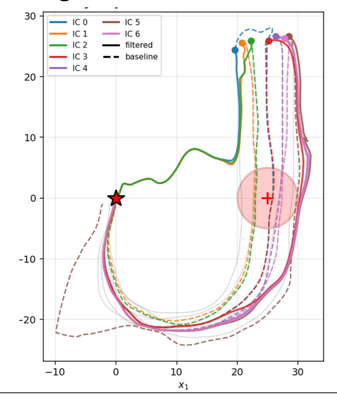








Safety Behavior using HJB



References

[1] On the Forward Invariance of Neural ODEs - Xiao, Wang, Rus et al.
[2] Learning Stable Deep Dynamics Models - Manek, Kolter
[3] A Forward Reachability Perspective on Control Barrier Functions and Discount Factors in Reachability Analysis - Choi, Lee, Tomlin et al.
[4] A Physically-Consistent Bayesian Non-Parametric Mixture Model for Dynamical System Learning – Figueroa, Billard